

HOME DEPARTMENT · GOVERNMENT OF GUJARAT

PRAHARI

Unified CCTV Intelligence Grid

Hybrid architecture: Model 1 registry + GIS · Model 2 unified viewing & ANPR · thin Model 3 event bus
Model 4 central VMS is Phase-2 — not a rip-and-replace of 26 departmental estates.

Student category · BITS Pilani · Team: Lead + Arnav + Aria · Submission 07 Sep 2026

The problem as deployed, not as a slide

26 departments

Independent VMS, mixed analog/IP, cloud and local NVR, 7-day vs 15-day retention.

~1,000 km span

Valsad to Palanpur, Dahod to Dwarka. A single central recorder is the expensive answer.

Watchlists already exist

VAHAN, SARTHI, eGujCop, AFIS, NAFIS — unused by the cameras that could match them.

No statewide census

Nobody can see which public-domain cameras exist, whether they are alive, or where the gaps are.

Why hybrid — and why not Model 4 first

Reference	PoC stance	Role
Model 1	YES — now	Census, GIS, health, gap analysis. Zero disturbance.
Model 2	YES — now	Direct RTSP / HLS / WHEP / ONVIF. Wins the test case.
Model 3	Thin bus	Detection events, watchlist match, cross-camera track.
Model 4	Phase-2	Central recording only for selected cameras. Not 80k × 15 days on day one.

End-to-end architecture



Open contracts. A new vendor is one adapter. Kafka replaces the PoC WebSocket without changing the detection JSON. Model 4 recording sits behind the same ingest, on selected cameras only.

Evaluation workflow

1. Onboard Sentinel RTSP/HLS IDs into the registry
2. Sample a frame · localise plate · OCR · normalise to GJ01AB1234 form
3. O(1) watchlist match against VAHAN / eGujCop-style table
4. CRITICAL alert + crop + camera GIS point
5. GET /api/track/{plate} rebuilds the statewide path and CSV report

AI approach — ANPR first, because that is the test

PoC engine

OpenCV morphology localiser + Tesseract.
Indian-plate regex gate. Operator-confirm
fallback so a visible plate never fails the
demo.

Production swap

Same recognize() contract. YOLOv8/11 plate
detector + PaddleOCR or a certified ANPR
SDK. No API change.

What we refuse to fake

No face-recognition theatre. FRS only later,
lawful watchlist, human confirm, dedicated
cameras.

Timestamp rule from the official Integrator Guide: never use packet arrival time; never trust CAP_PROP_FPS. Event time = PTS or server ingest clock.

Bonus analytics if GPU lands before 7 Sep: vehicle count on traffic nodes, intrusion on Food & Civil Supplies godowns. Not on the critical path.

Watchlist correlation and alert path

01

Detect

Plate event with camera, lat/lon, crop, confidence.

02

Match

In-memory set refreshed every 60s from VAHAN / eGujCop replicas.

03

Prioritise

CRITICAL stolen/wanted · HIGH blacklist · LOW observe.

04

Act

SOC queue, ack, audit. Dedupe 120s same plate+camera.

PoC ships with a representative watchlist (stolen, blacklist, missing-person associate, interstate plate). Production adapters read VAHAN / SARTHI / eGujCop / AFIS-NAFIS. The matcher does not care which side loaded the row.

Demo plate GJ01AB1234 is pre-seeded as STOLEN so the evaluation reconstruction is deterministic, then live detections append to the same path.

Test case — reconstruct a designated vehicle

Seeded evaluation path for GJ01AB1234 (live sandbox hits append automatically)

	06:12	Valsad NH-48 Toll	CAM-VAL-001
	07:48	Surat Adajan Circle	CAM-SUR-001
	09:05	Narol–Naroda Hwy	CAM-AHD-004
	09:22	SG Highway Junction	CAM-AHD-001
	09:51	Koba Circle	CAM-GNR-003
	10:04	CH-0 Sector 21 GNR	CAM-GNR-001

80,000 cameras without shipping 80,000 videos

45k

Public-domain + checkpoint cameras
run analytics

1 fps

Sampled frame, not 25 fps GOP, for
ANPR

5 regions

Ahmedabad · Surat · Rajkot ·
Vadodara · Bhuj

~5–6 Cr

Indicative yearly cost of the
intelligence plane, not a VMS rip-out

SOC video wall pulls 16–64 HLS/WebRTC tiles. Everything else is metadata + crops. Low-connectivity outposts (Dahod, border) run the worker on the NVR and send events only.

Hot 7 days / warm 30 / cold 90 — and only on cameras that policy marks for recording. That is how Model 4 arrives without bankrupting storage.

Security, privacy, audit

RBAC

superadmin · soc_operator · dept_viewer · auditor. Department viewers cannot open another estate.

Streams

RTSP never exposed to the public internet. Browser tiles use short-lived tokens on HLS/WebRTC.

Data

Centre stores events and watchlist crops. Full video stays with the owning department unless Phase-2 recording is on.

Audit

Onboard, watchlist edit, alert ack, report download — append-only log.

Network

Camera VLAN ≠ analytics VLAN ≠ enterprise VLAN. mTLS between regional workers and the bus.

Legal

Private mall / society feeds require a consent flag. No feed, no pin.

Stack — open, swappable, already running

UI

React-ready · current PoC is Leaflet + vanilla JS command centre

API

FastAPI · REST + WebSocket. Postgres/PostGIS in production, SQLite in the laptop PoC

Ingest

FFmpeg · OpenCV · official RTSP-TCP / HLS / WHEP endpoints

AI

OpenCV + Tesseract now · YOLO + PaddleOCR drop-in

Bus

WebSocket now · Redis / Kafka statewide

Ops

Health endpoint · camera SLO · Docker Compose → later Kubernetes

What changes for a duty officer on night shift

Minutes, not hours

A stolen plate lights up the next camera instead of a CD being couriered across districts.

One glass

Home, RTO, Food & Civil Supplies, GSRTC and permitted mall feeds on one map.

Honest gaps

Dahod offline, short-retention godowns, missing highway coverage — visible to planners.

Keep the AMC

Departments keep vendors and storage. The state buys an intelligence plane, not a second VMS bill.

What the working platform already shows

GIS registry of 25 cameras across Home, RTO, Food & Civil Supplies, GSRTC and private-permitted malls

Live ANPR on an uploaded still or an RTSP/HLS grab

Health layer and gap report (Dahod offline, short retention)

Operator-confirmed sighting so a readable plate cannot fail evaluation

Watchlist with VAHAN / eGujCop-style categories

Alert queue with priority and ack

Deterministic multi-camera track + CSV for GJ01AB1234

Onboarding form + CSV + REST — ready for the official ~50 sandbox IDs

Ask of the committee

Shortlist PRAHARI for the 10–11 September on-site PoC.

We will onboard the official sandbox cameras, lock ANPR on the designated plate, and run watchlist alerts against a live stream — on hardware we will bring to iHub.

Working tree: [prahari/](#) · [HLD: docs/HLD.md](#) · [War-room tracker: docs/PRAHARI_WarRoom_Tracker.xlsx](#)